



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch: B.E.VI , Computer Science and Engineering

Course Code & Title: CS8075 - Data Warehousing And Data Mining

Name of the Faculty member (s): Mrs.M.Swarna Sudha

Innovative Practice Description

Unit III -Topic: Association rule mining in data mining

Course Outcome: CO3

Topic Learning Outcome: TL07

Date of the activity : 05.05.2022

Activity Chosen: Problem-based learning- Collaborative learning

Justification:

Association rule in data mining is an important concept in which students should have depth knowledge than they will be able to solve the problems. The association rule learning is the significant technique of machine learning, and it is employed in Market Basket analysis, Web usage mining, continuous production, etc. In market basket analysis, it is an adequate used by several big retailers to find the relations among items. The students should have in-depth understanding in association rule concept then only they will be able to apply to the solution to several big retailers to find the relations among items

Time Allotted for the Activity: 25 minutes

Details of the Implementation:

- Allowed the students to make the group on own.
- Individual problem statements related to the association concepts were given to the students prior to the start of the event for preparation purposes.
- All the students actively shared their views in the group to solve the problems given to the them.
- Finally, faculty member consolidated the key concepts used in the solving the problems that was discussed in this activity.
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CO – PO / PSO mapping:

CO	PO1	PO2	PO9	PO10	PSO3
CO1	2	3	3	3	3

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO9	PO10	PSO3
	2	3	3	3	3
Justification for correlation	Apply the knowledge of data mining concepts to solve problems.	Identify the solution to the problems and making rule the association related to the problems	Each one sharing the ideas individual land working in team to find the solution to solve the problems	Effective communication involves exchanging ideas and encouraging students to present their own ideas.	Gaining the knowledge in Data analytics for generating the rule to Data mining problems

Screenshot of the practice:

Figure 1 : Students are involved in group and working together to solve problems.

**Reflective Critique:**

- Students' ability to solve problems will be increased.
- It help the students to understand the concepts in depth
- It also help the students to view the problems in different aspects since each one of them gave their ideas to solve the problems.

Benefit of the practice:

- Students contribute with enthusiasm, and this exercise encouraged them to impart their knowledge to others.
- By debating and exchanging ideas with their classmates during this activity, the students can gain a better understanding of association rule problems.

- Students are able to describe concepts effectively.

Which one of the following is alternative search strategies for mining multiple-level associations with reduced support?

Level – cross-filtering by a single item		0 %	
Level – by level independent	1 respondent	3 %	
Level – cross-filtering by k – itemset:		0 %	
All	35 respondents	95 %	 ✓
No Answer	1 respondent	3 %	

Fig 2 In Unit III quiz 95% of the students have answer correctly

Challenges faced in implementation:

- Students believe that learning is a time-consuming task.
- It is difficult for every student to contribute equally.
- Always the toppers only represent the concepts
- Some students feel the discussion is challenging because they lack the confidence.

References:

- <https://educationaltechnology.net/problem-based-learning-pbl/>
- <https://study.com/learn/lesson/problem-based-learning-overview-examples.html>
- <https://www.teachthought.com/pedagogy/20-collaborative-learning-tips-and-strategies/>
- https://www.ritrjpm.ac.in/images/computerscience/28.CS8501_ColloborativeLearning.pdf

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